MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 8922 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Passive TR-limiter, crystal protector, ignitorless, 2,600 to 3,100 MHz

ABSOLUTE RATINGS:

Parameter:	Incident power	Du	tp	Pressurization	Temperature Operating	Temperature Nonoperating	Altitude
Unit:	kw	---	μs	psia	°C	°C	ft
Maximum:	50	0.03	36	40	+65	+75	30,000
Minimum:	0	---	---	---	0	-62	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Mounting:	Input and output flanges mate with M3922/61-1
Mounting position:	Any		
Cooling:	Convection	Flanges:	See figure 1 (note a)
Finish:	See figure 1 (note b)		

TEST CONDITIONS:

Parameter:	Incident power	Du	tp	F
Unit:	kw	---	μs	MHz
Tolerance:	±10%	min	±0.10	±3%
	30	0.007	6.0	F3

FREQUENCY		
F	MHz	±
1	2,600	0.1%
2	2,900	0.1%
3	3,000	0.1%
4	3,100	0.1%

GENERAL:

First article testing - Required (see note 1)
Service life guarantee - See notes 2 and 3
Shelf life - See note 3

1/ Replaces type GIB 182858-1

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	4					
4473	Low-level VSWR	-	F = F2 thru F4	σ	---	1.35	---
4416	Insertion loss	-	F = F2 thru F4	Li	---	0.4	dB
4452	Flat-leakage power	-		pf	---	60	mw
4452	Spike-leakage energy	-		Ws	---	0.15	erg
4471	Recovery time	-		t	---	20	μs
4496	Firing power	-		Po/Du	---	200	mw
1027	Temperature cycling (nonoperating)	5	One cycle	---	---	---	---
	Quality conformance inspection, part 2						
---	Noise ratio	6		Nr	---	1.05	---
4003	Pressurizing	5, 8	40 psia; 1 cycles	---	---	---	---
4101	Dielectric material strain	7		---	---	---	---
1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	2, 3	Group S	t	2,000	---	hrs
---	Life-test end points:						
---	Crystal degradation	9, 10		ΔNF	---	2.0	dB
4416	Insertion loss	-	F = F2 thru F4	Li	---	0.5	dB
4452	Spike-leakage energy	-		Ws	---	0.3	erg
4471	Recovery time	-		t	---	30	μs
4496	Firing power	-		Po/Du	---	600	mw
	Periodic-check tests						
4003	Pressurizing	5, 8	40 psia; 10 cycles	---	---	---	---
1042	Shock, specified pulse	11	Test condition A	---	---	---	---
1032	Vibration, mechanical	12		---	---	---	---
1027	Temperature cycling life test	5	40 psia; 10 cycles	---	---	---	---
---	Humidity	13		---	---	---	---
---	Performance degradation	14		---	---	---	---

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NOTES:

1. First article testing requirements hereby replace any qualification requirements referable to the product covered herein. The term "First article testing" shall be considered as synonomous with the term "Preproduction sample approval testing". All tests applicable herein (including all First article tests and quality conformance inspection, parts 1, 2, and 3) shall be performed during first article testing. A failure of any one tube in any of the tests shall be cause for decision of first article sample disapproval. The contractor/manufacturer shall provide test reports and written assurance that:

   (a) The first article sample is representative of the product to be submitted under the contract.

   (b) First articles shall be manufactured from new materials and in the facility that produced the sample tubes.

   A test plan shall be submitted to the contracting officer for approval prior to proceeding with first article testing. The test plan shall list all production test and laboratory test facilities to be utilized in first article testing and production phases of the contract. Upon receipt of approval of the test plan, first article testing may proceed. When testing has been completed, three copies of the test report shall be prepared in accordance with MIL-STD-831, certified by a government representative and submitted with the test samples to the contracting officer for approval.

2. Life tests shall not be required after the TUT have satisfied all requirements of this specification for first article approval. However, a service-life guarantee of 2,000 hours minimum shall be required in accordance with the requirements of MIL-E-1. Life for purpose of the service-life guarantee shall be defined as the total rf operating time. The sum of the operating and shelf life shall be not less than 21 months.

3. The tubes tested and procured in accordance with this tube specification sheet shall be capable of shelf-life for a period of 18 months after date of shipment from the manufacturer's plant. Tubes placed in operation during this shelf-life period shall meet all requirements of this tube specification sheet. Tubes found to be defective within the shelf-life period shall be replaced by the manufacturer at no cost to the Government or its contractors.

4. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

5. After the completion of this test, the TUT shall satisfy all requirements of method 4452 (spike-leakage energy), as specified in quality conformance inspection, part 1.

6. The noise ratio shall be defined as the ratio of the IF attenuation required to reduce the output noise level with the TUT installed to that value when the TUT is removed from the circuit. The noise ratio can be determined from the following equation:

$$Nr = 1 + \frac{\text{antilog } (0.1F) \quad [\text{antilog } (0.1Y) - 1]}{2}$$

   Where F = Receiver noise figure in dB
   Y = IF attenuation required to reduce the output noise level with the TUT installed in the circuit to that with the TUT removed from the circuit.

   Note: The denominator in the second term is applicable only if no preselector is used.

7. After the completion of this test, the TUT shall satisfy all requirements of method 4471 (recovery time), as specified in quality conformance inspection, part 1.

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NOTES: -Continued

8. The specified air pressure shall be applied to the input window of the TUT for a period of 30 minutes, after which the pressure shall be reduced to atmospheric pressure. This test shall be performed by immersing the seal and tube flange in water. The maximum allowable leak rate shall be one bubble per 30 seconds. Bubbles due to occluded air shall be disregarded. This test shall be performed for the number of cycles specified.

9. The tube shall be terminated with a type 1N21E crystal mounted in a crystal holder and be provided with a dc ground return of 100 ±10 ohms. At the beginning of the life test, the crystal noise figure (NF) shall meet the minimum requirements for the type 1N21E crystal under the standard specified conditions. The initial crystal noise figure shall be measured and recorded. The crystal noise figure shall be remeasured periodically during the course of life testing. Life test shall be considered completed when the 1N21E crystal noise figure degradation (ΔNF) from the initial value has exceeded the specified limit.

10. One crystal failure will be allowed during the life test. If a crystal failure occurs, a new crystal shall be inserted in the mount and the life test shall continue for a minimum of 200 hours, even if the life test is extended beyond the required 2,000 hours.

11. The TUT shall be shocked 5 times in the direction shown in figure 1 when actual mounting conditions are simulated. The TUT shall be capable of satisfying all requirements of this specification without damage or failure before, during, and after being subjected to the specified shock test.

12. Vibration. The TUT shall be capable of satisfying all requirements of this specification without damage or failure before, during, and after being subjected to the specified vibration test with the amplitude as specified herein.

Frequency (Hz)		Amplitude (Inches)
5 to 15	0.075 ±.012	(0.150 Total displacement)
16 to 25	0.050 ±.008	(0.100 Total displacement)
26 to 33	0.025 ±.004	(0.050 Total displacement)

13. The TUT shall perform as specified herein without degradation of performance when its operating or nonoperating humidity environment has a relative humidity up to 95 percent, including condensation at any temperature up to 50°C (122°F).

14. The TUT shall not exhibit damage or degradation of performance resulting from spurious signals within F1 through F2 and incident power = 5 kw maximum.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU, MI
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-T034)

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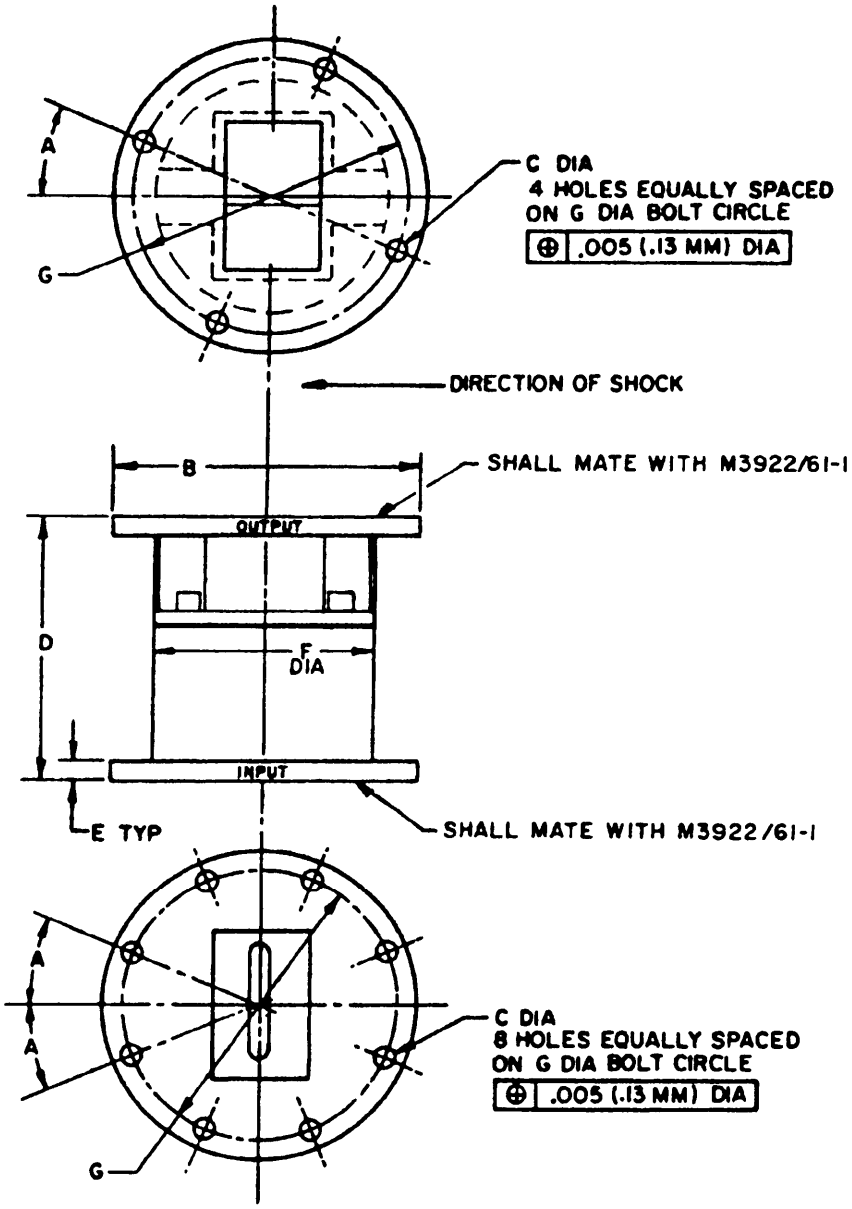


FIGURE 1. Outline drawing of electron tube type 8922.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Preproduction sample approval inspection		
A	22°	23°
E	.235 (5.97)	.265 (6.73)
F		4.000(101.60)
Quality conformance inspection, part 1		
C	.256 (6.50)	.261 (6.63)
D	3.469(88.11)	3.531 (89.69)
Quality conformance inspection, part 2		
B	5.297(134.54)	5.327(135.31)
G	4.745(120.52)	4.755(120.78)

NOTES:
a. The waveguide flanges of the TR tube shall be cold rolled steel.
b. All external and interface surfaces of the TR tube, with the exception of the aluminum parts, shall be cadmium plated in accordance with Federal Specification QQ-P-416, type II, class 1. Aluminum parts shall have a chemical film finish in accordance with MIL-C-5541, dated 9 January 1950. All external surfaces, with the exception of the mating surfaces of the waveguide flanges shall be wash primed in accordance with MIL-C-8507 one coat .0002(.005 mm) to .0005 .013 mm) thick with primer in accordance with MIL-P-15328. Paint two coats .001(.03 mm) thick with gray enamel in accordance with MIL-E-15090, type II, class 2.
c. The rf input and output shall be marked as indicated.
d. The rf input and output flanges shall mate with M3922/61-1.

FIGURE 1. Outline drawing of electron tube type 8922 - Continued.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 66 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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NAVAL ELECTRONIC SYSTEMS COMMAND
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